AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

MT131
Discrete Mathematics

Final Examination
Summer Semester 2024/2025

Date: 02 /Sept./2025

Number of Exam Pages: 3 (including this cover sheet)	**Time Allowed:** 2 hours

Instructions:

- Total Marks: **50**
- This exam consists of **5** questions.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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Q–1: [5×2 marks] Choose the correct answer:

1.1: Using c for "It is cold" and d for "It is dry", the statement "It is neither cold nor dry" in symbols is

a) $\neg c \wedge d$
b) $\neg(c \vee d)$
c) $\neg(c \wedge d)$
d) $\neg c \rightarrow \neg d$
e) None of the above

1.2: The negation of the statement $x \notin A$ or $x \in B$ is:

a) $x \in (A - B)$
b) $x \in (B - A)$
c) $x \notin A$ or $x \in B$
d) $x \in (A \cap B)$
e) none of the above

1.3: The decimal notation for the number 1 0 0 1 1 in binary notation is :

a) 18
b) 19
c) 35
d) 65
e) None of the above

1.4: Let $R = \{(a,a),(a,b),(a,c),(b,a),(b,b),(b,c),(c,a),(c,c)\}$ be a relation on the set $A = \{a,b,c\}$. The reflexive closure of R is

a) $\{(a,a),(a,b),(a,c),(b,a),(b,c),(c,a),(c,c)\}$
b) R
c) $\{(a,a),(b,b),(c,c)\}$
d) $\{(a,a),(a,b),(a,c),(b,a),(b,b),(b,c),(c,a)\}$
e) None of the above

1.5: The number of edges of the complete bipartite graph $K_{2,7}$ is

a) 9
b) 7
c) 2
d) 14
e) None of the above

Each question is worth 10 marks. Answer the following questions:

Q–2:

a) **[5 marks]** Using truth tables, determine whether the two propositions $p \rightarrow (\neg q \wedge r)$ and $\neg p \vee \neg(r \rightarrow q)$ are logically equivalent.

b) **[3 marks]** Prove that $\neg\,(p \rightarrow (q \rightarrow p))$ is a contradiction, by using law of logic.

c) **[2 marks]** Let $g(n) = \max\{0, n-3\}$ (e.g. g(1)=0 and g(5)=2) and Let $A = \{0,2,4,6\}$

i. Find $g(A)$.
ii. Find $g^{-1}(5)$.

Q–3:

a) **[2+2+1 marks]** Suppose a license plate consists of 3 letters followed by 3 digits. Letters are chosen from the 26-letter English alphabet, and digits from 0 to 9. Repetition is allowed.

i. How many total license plates are possible?
ii. How many license plates end with the digit 7?
iii. How many license plates begin with the letter R or T and end with the digit 7 or 8?

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b) **[2+3 marks]** Suppose a factory produces items where each item has a probability of 0.25 of being defective and 0.75 of being non-defective. A random sample of 10 items is chosen. Find:

i. P(exactly 7 non-defective items)

ii. P(at least 7 non-defective items)

Q–4:

a) Let R be the relation on the set $A = \{1,2,3,4\}$ defined by xRy if and only if |y-x|=1 (i.e, $-1 \leq x - y \leq 1$)

i. **[2 marks]** List the members of R.

ii. **[2 marks]** Write the matrix representing R.

iii. **[2 marks]** State whether R is reflexive, symmetric, transitive and/or antisymmetric.

iv. **[2 marks])** Find the symmetric and reflexive closures of R.

b) **[2 marks]** Let R be a partial order relation on $A = \{1,3,5,10,15,20\}$ where aRb means $a|b$. Draw the Hasse diagram for the relation R.

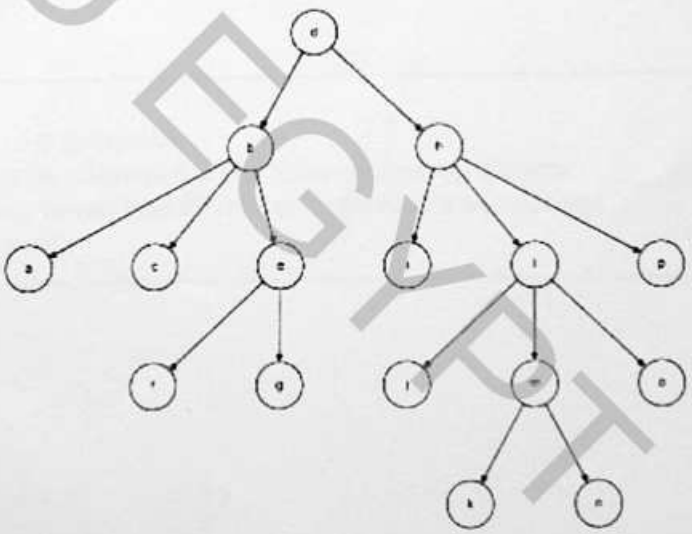
Q–5:

a) Let $A = \begin{bmatrix} 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 \\ 1 & 1 & 0 & 0 & 0 \end{bmatrix}$ be the incidence matrix of a graph G.

i. **(3 marks)** Draw G.

ii. **(4 marks)** Find the adjacency matrix of G.

b) **[3 marks]** Given the tree:



i. What is the height of the tree?

ii. What is the level of *g*?

iii. Determine whether the tree is a balanced.

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End of questions

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